

Daniel DeAlcala | PhD Student,

Universidad Autónoma de Madrid

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Education

Ph.D., in Telecommunications and Computer Engineering **Madrid, Spain**
Universidad Autonoma de Madrid (UAM) *Jan 2022 – Ongoing*

- **Funded by a FPU Fellowship** (Spanish National Programme for PhD Training) one of the most competitive national PhD fellowships in Spain.
- **Supervisors:** Julian Fierrez and Aythami Morales.

M.Sc., in Deep Learning for Audio and Video Signal Processing **Madrid, Spain**
Universidad Autonoma de Madrid (UAM) *Sept 2020 – Sept 2021*

- **Final Project:** *Application for the demonstration of the automatic registration of transited spaces for contact tracing of infectious diseases using video signals from life-logging cameras.* Advancing ego-centric video localization and multi-user graph modeling to enable accurate contact detection and infectious disease tracing in real environments.

B.Sc., in Telecommunications Engineering **Madrid, Spain**
Universidad Autonoma de Madrid (UAM) *Sep 2016 – Jun 2020*

- **Final Project:** *Egocentric Biometric Palmprint Recognition for Virtual Reality Environments.* Creating a complete biometric palm recognition system and dataset tailored for VR environments to enable future research and secure user authentication.
- **Ranked 1st** in class.
- EPS Effort Award and Excellence Scholarship.
- 8 courses with High Distinction.

R&D Projects Participation

- BBforTAI through MICINN/FEDER under Grant PID2021-127641OB-I00.
- HumanCAIC through MICINN under Grant TED2021-131787B-I00.
- M2RAI through MICIU/FEDER under Grant PID2024-160053OB-I00.
- Cátedra ENIA UAM-VERIDAS (NextGenerationEU) under Grant TSI-100927-2023-2.

International Research Stays

Research Intern **New York, USA**
IBM Research *Sept – Dec 2025*

- Research on **AI-Generated Content Detection** at document level.
- Focus on developing methods to identify, classify, and segment components of AI-generated documents, including text, images, and tables.
- Collaboration with IBM Research on state-of-the-art detection techniques and data creation.

Academic Duties

Conference Reviewer.....

- Computer Vision and Pattern Recognition (CVPR), 2023,2024,2025.
- Computer Vision and Pattern Recognition Workshops (CVPR Workshops), 2024,2025.
- Association for the Advancement of Artificial Intelligence Workshops (AAAI), 2025.
- Neural Information Processing Systems (NeurIPS), 2025.
- International Conference on Computer Vision (ICCV), 2025.

Teaching.....

- Multimedia Signal Processing, B.Sc. in Telecommunications Engineering (2023 - Present)

Publications

- [1] R. Daza, D. DeAlcala, A. Morales, R. Tolosana, R. Cobos, and J. Fierrez, "ALEBk: Feasibility Study of Attention Level Estimation via Blink Detection applied to e-Learning," *AAAI Workshop on Artificial Intelligence for Education (AI4EDU)*, 2022.
- [2] I. Serna, D. DeAlcala, A. Morales, J. Fierrez, and J. Ortega-Garcia, "IFBiD: Inference-Free Bias Detection," *AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, 2022.
- [3] D. DeAlcala et al., "BeCAPTCHA-Type: Biometric Keystroke Data Generation for Improved Bot Detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR)*, 2023, pp. 1051–1060.
- [4] D. DeAlcala, I. Serna, A. Morales, J. Fierrez, and J. Ortega-Garcia, "Measuring Bias in AI Models: An Statistical Approach Introducing N-Sigma," in *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, IEEE, 2023, pp. 1167–1172.
- [5] D. DeAlcala, G. Mancera, A. Morales, J. Fierrez, R. Tolosana, and J. Ortega-Garcia, "A Comprehensive Analysis of Factors Impacting Membership Inference," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR)*, 2024, pp. 3585–3593.
- [6] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, and J. Ortega-Garcia, "Is My Data in Your AI? Membership Inference Test (MINT) applied to Face Biometrics," *IEEE Access*, 2025.
- [7] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, and R. Vera-Rodriguez, "MINT-Demo: Membership Inference Test Demonstrator," *AAAI Workshop on Artificial Intelligence Governance (AIGOV)*, 2025.
- [8] G. Mancera, D. DeAlcala, J. Fierrez, R. Tolosana, and A. Morales, "Is My Text in Your AI Model? Gradient-based Membership Inference Test applied to LLMs," *arXiv preprint arXiv:2503.07384*, 2025.
- [9] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, and R. Tolosana, "gMINT: Gradient-based Membership Inference Test applied to Image Models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR)*, 2025, pp. 2781–2790.
- [10] D. DeAlcala, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, and J. Ortega-Garcia, "Active Membership Inference Test (aMINT): Enhancing Model Auditability with Multi-Task Learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 647–656.
- [11] D. DeAlcala, A. Morales, J. Fierrez, and R. Tolosana, "AttZoom: Attention Zoom for Better Visual Features," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCV)*, 2025, pp. 4834–4843.
- [12] P. Korshunov et al., "DeepID Challenge of Detecting Synthetic Manipulations in ID Documents," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 510–519.
- [13] D. DeAlcala, K. Ching-Yun, A. Morales, J. Fierrez, R. Tolosana, and P.-Y. Chen, "AIDoc: A Framework for Generating and Identifying AI-Edited Documents," in *Under Review in International Conference on Machine Learning (ICML)*, 2026.

Participation to Conferences

- Computer Vision and Pattern Recognition Conference (CVPR), 2023. Vancouver, Canada.
- Computers, Software, and Applications Conference (COMPSAC), 2023. Torino, Italy.
- Computer Vision and Pattern Recognition Conference (CVPR), 2024. Seattle, WA, USA.
- The Association for the Advancement of Artificial Intelligence (AAAI), 2025. Philadelphia, Pennsylvania, USA.
- Computer Vision and Pattern Recognition Conference (CVPR), 2025. Nashville, TN, USA.
- International Conference on Computer Vision (ICCV), 2025. Honolulu, Hawaii.